Use Cases – Sports Event Ticketing System

Sample Use Case

Use case name:	User Registers and Logs into the System	**ID:**	1	**Importance level:**	High
Primary actor:	Fan (User)				
Short description:	This use case describes how a new user registers an account on the Sports Event Ticketing System and how an existing user logs in to access ticketing features.				
Trigger:	User visits the platform and chooses to register a new account or log in to an existing one.				
Type:	External				

Major Inputs

Description	Source
Full name	User
Email address	User
Phone number	User
Password	User
National ID	User
Login credentials (email + password)	User

Major Outputs

Description	Destination
Registration confirmation message	User (Email / SMS)
User account record	Database
Authentication token (session)	System / User
Login success / failure message	User
Access to user dashboard	User
Error notification (invalid credentials)	User

Major Steps Performed	→	Information for Steps
1. User opens the web or mobile application.	→	Platform URL / App
2. User selects "Register" to create a new account.	→	Register action
3. User fills in: full name, email, phone, national ID, and password.	→	Full name · Email · Phone · National ID · Password
4. System validates data format and checks for duplicate accounts.	→	User record (DB) · Validation rules
↳ *If invalid data → System shows error and prompts user to correct it.*	↩	Error message
5. System saves new user record and sends confirmation email / SMS.	→	User account record · Confirmation msg
6. User selects "Login" and enters email and password.	→	Login credentials
7. System authenticates credentials against the database.	→	User record (DB) · Hashed password
↳ *If credentials incorrect → System displays login failure message.*	↩	Failure message
8. System generates auth token and grants access to the user dashboard.	→	Auth token · Dashboard access

Sample Use Case

Use case name:	User Searches and Browses Available Matches	**ID:**	2	**Importance level:**	High
Primary actor:	Fan (User)				
Short description:	This use case describes how a logged-in user searches for available sports matches using filters such as sport type, date, venue, or team, and views real-time seat availability.				
Trigger:	User logs in and navigates to the "Browse Matches" section to find a specific event.				
Type:	External				

Major Inputs

Description	Source
Sport type filter	User
Match date filter	User
Venue / stadium filter	User
Team name filter	User
Available matches list	Database
Real-time seat availability data	Database / System

Major Outputs

Description	Destination
Filtered list of matches	User
Match details (date, time, teams, venue)	User
Interactive seat map	User
Available / unavailable seats	User
Ticket price per seat category	User
No results message (if none found)	User

Major Steps Performed	→	Information for Steps
1. User logs in and navigates to "Browse Matches".	→	Auth token · Browse Matches page
2. System displays all upcoming available matches from the database.	→	Available matches list
3. User applies filters: sport type, date, venue, and/or team name.	→	Sport type · Match date · Venue · Team name
4. System queries the database with applied filters and returns results.	→	Filtered matches list
↳ *If no matches found → System displays "No available matches" message.*	↩	No results message
5. User selects a specific match to view full details.	→	Match details
6. System displays date, time, teams, venue, and ticket pricing.	→	Match details · Ticket price categories
7. System loads interactive seat map with real-time availability. Green = available | Red = sold / unavailable	→	Seat map data · Seat status (DB)
8. User reviews seat map and pricing, then proceeds to booking if desired.	→	Selected match · Available seats

Sample Use Case

Use case name:	User Books a Match Ticket	**ID:**	3	**Importance level:**	High
Primary actor:	Fan (User)				
Short description:	This use case describes how a logged-in user selects a seat for a specific match, proceeds through the payment process, and receives a booking confirmation. Ticket sales open 20 days before a match and close 3 hours before it starts.				
Trigger:	User selects an available seat from the seat map and clicks "Book Now".				
Type:	External Temporal				

Major Inputs

Description	Source
Selected match	User
Selected seat(s)	User / Seat map
Ticket quantity	User
Payment method (Fawry / Paymob)	User
Payment details	User
Booking window status (open / closed)	System

Major Outputs

Description	Destination
Seat reservation (temporary hold)	Database / Seat map
Payment request	Payment gateway
Payment confirmation	User / System
Booking confirmation record	Database
Booking confirmation notification	User (Email / App)
Payment failure / booking closed message	User

Major Steps Performed	→	Information for Steps
1. User selects an available match and views the seat map.	→	Selected match · Seat map
2. System checks if the booking window is open (opens 20 days before, closes 3 hours before match).	→	Booking window status
↳ *If booking is closed → System displays "Booking unavailable" message.*	↩	Closed booking msg
3. User selects seat(s) and specifies ticket quantity.	→	Selected seat(s) · Ticket quantity
4. System temporarily reserves selected seats to prevent overselling.	→	Seat reservation (temp hold)
5. System displays booking summary (match, seat(s), total price).	→	Booking summary · Total price
6. User selects payment method (Fawry or Paymob) and enters payment details.	→	Payment method · Payment details
7. System sends payment request to selected payment gateway.	→	Payment request · Gateway response
↳ *If payment fails → System releases seat hold and notifies user.*	↩	Payment failure msg · Seat released
8. System confirms payment, finalizes booking, and saves record in database.	→	Payment confirmation · Booking record
9. System sends booking confirmation to user via email and/or in-app wallet.	→	Confirmation notification · Email / App wallet

Sample Use Case

Use case name:	System Generates and Delivers QR Code Ticket	**ID:**	4	**Importance level:**	High
Primary actor:	System (Automated)				
Short description:	This use case describes how the system automatically generates a unique, encrypted, scannable QR code ticket after a successful booking payment, and delivers it to the user via email and the in-app wallet.				
Trigger:	Payment is confirmed successfully and a booking record is saved in the database.				
Type:	External Temporal				

Major Inputs

Description	Source
Confirmed booking record	Database
User information (name, email)	Database
Match details (date, venue, seat number)	Database
Payment confirmation reference	Payment gateway
Unique ticket ID	System (generated)
Encryption key	System

Major Outputs

Description	Destination
Unique encrypted QR code	System / User
QR code ticket stored in database	Database
Ticket delivered via email	User (Email)
Ticket added to in-app wallet	User (App wallet)
Delivery confirmation log	System log
Delivery failure alert (if fails)	System / Admin

Major Steps Performed	→	Information for Steps
1. System detects successful payment confirmation from the payment gateway.	→	Payment confirmation ref
2. System retrieves the confirmed booking record (user info, match details, seat number).	→	Booking record · User name & email · Match & seat details
3. System generates a unique ticket ID for the booking.	→	Unique ticket ID
4. System creates an encrypted QR code embedding ticket ID, match details, seat, and user identity.	→	Encrypted QR code · Encryption key
5. System stores the QR code ticket record in the database, linked to the booking.	→	QR ticket (DB record)
6. System sends the QR code ticket to the user via email.	→	Email delivery · User email
↳ *If email fails → System retries and logs failure; Admin is notified.*	↩	Failure alert · Retry log
7. System adds the QR code ticket to the user's in-app wallet.	→	In-app wallet entry
8. System logs successful delivery and updates booking status to "Ticket Issued".	→	Delivery confirmation log · Status: Ticket Issued

Sample Use Case

Use case name:	Admin Manages Events, Pricing, and Seat Allocation	**ID:**	5	**Importance level:**	High
Primary actor:	Admin (Ministry of Youth and Sports Staff)				
Short description:	This use case describes how an authorized admin logs into the admin dashboard to create, update, or remove sports events, manage seat allocation and ticket pricing, and generate reports on sales and attendance.				
Trigger:	Admin logs into the admin dashboard to add a new event, update an existing one, or review system reports.				
Type:	External				

Major Inputs

Description	Source
Admin credentials (username + password)	Admin
Event details (name, date, teams, venue)	Admin / Ministry
Seat categories and capacity	Admin
Ticket pricing per category	Admin
Event status (active / cancelled)	Admin
Report filter criteria (date range, event)	Admin

Major Outputs

Description	Destination
New / updated event record	Database
Published event on platform	Users / Platform
Updated seat map and pricing	Database / Users
Cancellation notifications (if event removed)	Users (Email / App)
Sales and attendance report	Admin
Operation success / error message	Admin

Major Steps Performed	→	Information for Steps
1. Admin navigates to the admin dashboard and logs in with credentials.	→	Admin credentials
2. System authenticates admin and grants access to the admin panel.	→	Auth token; Admin panel access
↳ *If credentials invalid → System displays error message.*	↩	Error message
3a. Add New Event: Admin enters event name, sport type, date, time, teams, and venue. Sets seat categories, capacity, and ticket prices.	→	Event details; Seat categories; Ticket pricing
↳ *System saves the new event and publishes it on the platform.*	→	New event record; Published event
3b. Update Existing Event: Admin selects an event and modifies required fields (date, pricing, seats).	→	Updated event record; Updated seat map
3c. Cancel Event: Admin marks event as cancelled.	→	Event status: Cancelled; Cancellation notification
↳ *System sends cancellation notifications to all ticket holders and initiates refund process.*	→	User notifications; Refund trigger
3d. View Reports: Admin selects report type and filters by date or event.	→	Report filter criteria
↳ *System generates and displays real-time sales, attendance, and revenue reports.*	→	Sales report; Attendance report
4. System confirms the operation and displays success message to Admin.	→	Success / error message